An Internship Report

Of

Data Base management

Submitted To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech (Computer Science and Engineering)

By

Roll Number of Student: 2410030562

Name of Student: Aditya Gupta

Batch: 2024-2028

CANDIDATE'S DECLARATION

I, ADITYA GUPTA, do hereby declare that the internship report titled "Database Management Internship at Acapella Productions" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Student Name: Aditya Gupta

Roll No.: 2410030562

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluably constructive criticism, and friendly advice during this work.

I would like to thank the team at Acapella Productions for providing me the opportunity and the structured platform to learn and gain hands-on exposure in the domain of relational database management and MySQL optimization. I am also thankful to my institute, [Your University Name], and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

Student Name: Aditya Gupta **Roll No.:** 2410030562

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3. INTERNSHIP COMPLETION CERTIFICATE

ACAPELLA

Acapella Productions

C-1/B MANSAROVAR GARDEN NEW DELHI-110015, TEL. 8447858687

EMAIL - Info@acapellaproductions.com, jasvinder4522@gmail.com

Ref: Acapella/EL/26-27/091

Date: 28.08.2026

Sub: Experience Letter

To whomsoever it may concern

This is to certify that **Aditya Gupta** has successfully completed his internship with **Acapella Productions** as a **Database Management Intern** from **May 01, 2026 to July 31, 2026.**

During the internship period, Aditya demonstrated exceptional technical skills, professionalism, and a strong aptitude for problem-solving and innovation. He was primarily responsible for assistance in managing database records, optimizing queries, and assisting with data organization, proving to be efficient, reliable, and highly effective in achieving the intended goals.

During this tenure, we found him very sincere, creative, and hardworking. This is to confirm that he has been relieved from all his responsibilities and holds no dues towards the company.

We take this opportunity to thank him and wish him success in his future endeavours. Warm

regards,

For Acapella Productions

Proprietor

Authorized Signatory Acapella Productions

4. PROJECT DESCRIPTION

4.1 Introduction This report presents a detailed account of the 3-month summer internship undertaken at Acapella Productions in the domain of Database Management. The internship was carried out from May 2026 to July 2026 as part of the requirement of the Bachelor of Technology (Computer Science and Engineering) programme. The primary aim of this internship was to build a strong practical foundation in Database Management Systems (DBMS) using MySQL, and to apply it to real-world backend challenges such as query optimization, data normalization, and database structuring for event management logistics.

4.2 Organization Profile Acapella Productions is a premier event and talent management organization that orchestrates large-scale corporate events, sports events, live shows, college fests, and government events. The company's operations include comprehensive event production and talent management for high-profile artists. Managing the logistics for these operations requires highly scalable and robust database architecture to efficiently handle artist schedules, event metadata, client records, and vendor details seamlessly.

4.3 Problem Statement Although theoretical foundations of database management and SQL are taught during academic coursework, structured, hands-on exposure to live corporate databases is essential for practical mastery. The challenge addressed by this internship was to optimize and manage large datasets related to event logistics and talent schedules. Legacy tables often suffered from data redundancy and slow retrieval times during complex multi-table joins. The project required applying normalization techniques and indexing strategies to improve overall database performance and structural integrity.

4.4 Project Objectives

- To understand the existing relational database schema used by the backend team.
- To write, test, and optimize complex SQL queries for faster data retrieval.
- To apply database normalization rules (1NF, 2NF, 3NF) to restructure legacy data tables and eliminate redundancies.
- To assist in the secure migration of legacy talent and vendor data to updated relational schemas.
- To implement routine database backup protocols and relational constraints (primary/foreign keys).

4.5 Scope of the Project The scope of this internship spans the complete backend database management workflow using MySQL. It includes conducting data audits, restructuring tables for talent and event management, and optimizing query execution times. The internship was focused on the database layer and backend operations, collaborating with software developers to ensure seamless data integration into daily internal workflows.

4.6 Technologies and Tools Used

- **Core Language:** SQL (Structured Query Language)
- **RDBMS (Relational Database Management System):** MySQL
- **Database Tools:** MySQL Workbench, Command Line Interface (CLI)
- **Concepts:** Database Normalization, Query Optimization, Indexing, Relational Schema Design

4.7 System Architecture The general architecture followed during the internship can be summarized in the following layers:

- **Data Storage Layer:** The core MySQL relational database housing multiple tables for events, artists, clients, and vendors.
- **Schema & Constraint Layer:** Applied primary keys, foreign keys, and unique constraints to maintain strict data integrity and prevent orphaned records.
- **Query Optimization Layer:** Implementation of indexed views and optimized join operations to reduce server load and latency.
- **Application Integration Layer:** Serving the structured and optimized data to the backend application layer for end-user (staff) access.

4.8 Methodology The internship followed a structured, phase-wise methodology combining conceptual application and hands-on optimization, as summarized in the table below:

Phase	**Module**	**Description**
Phase 1 (May)	Database Auditing & Schema Familiarization	Analyzed the existing database architecture. Identified bottlenecks, redundancies, and missing constraints in legacy tables related to vendor and event data.
Phase 2 (June)	Database Normalization	Restructured existing tables to adhere to standard normal forms (up to 3NF). Separated mixed data into distinct relational tables connected by foreign keys.
Phase 3 (July)	Query Optimization & Indexing	Rewrote inefficient queries causing high latency. Implemented appropriate indexing on frequently searched columns to drastically reduce data retrieval times.
Phase 4 (Late July)	Data Migration & Documentation	Supported the transfer of legacy talent data into the updated schema. Documented database structures and backup protocols for the development team.

4.9 Expected Outcomes

- Practical understanding of managing live corporate databases at scale.
- Ability to apply database normalization techniques to eliminate structural redundancies.
- Hands-on expertise in identifying slow-performing SQL queries and optimizing them using indexing and execution plan analysis.
- Capability to collaborate with a technical team to ensure backend data integrity.
- Successful implementation of a streamlined data architecture for Acapella Productions' event logistics.

5. BIBLIOGRAPHY/REFERENCES

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